

**STEP 1**

# Environment setup

Create an isolated Python environment with exactly the libraries the project needs.

## What and why

Each project gets its own **virtual environment** (venv) so its library versions do not clash with other projects. scikit-learn is **pinned to 1.7.2** because a saved model must be loaded with the same version that trained it; a mismatch produces warnings or wrong results.

## Files

| File                 | Contents                                                                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| requirements.txt     | Runtime libraries: numpy, pandas, scikit-learn==1.7.2, scipy, joblib, matplotlib, fastapi, uvicorn, pydantic, streamlit, requests |
| requirements-dev.txt | Everything above plus pytest, httpx (API tests), jupyter, ipykernel                                                               |

## How to run

```
cd C:\Users\priya\MLOPS\food-waste-mlops
python -m venv venv
.\venv\Scripts\Activate.ps1          # prompt now starts with (venv)
pip install -r requirements-dev.txt

# make the venv available as a Jupyter kernel
python -m ipykernel install --user --name food-waste-venv --display-name "Python (food-waste venv)"
```

## Expected result

The terminal prompt shows (venv), and in Jupyter the kernel **Python (food-waste venv)** appears in the kernel list.

### Watch out

In Jupyter, always check the kernel name in the top-right corner. The default **Python 3** kernel uses the system Python (C:\Python310), which has different library versions. That caused the 'Trying to unpickle estimator from version 1.7.1' warning in the earlier IMDB project.

### Why it matters

Reproducibility starts here. Pinned versions plus a clean environment mean the model trained on your laptop, in CI and inside Docker is built from the same code with the same libraries.